

Lab 0: Getting Started with Datahub and R

POLI 172 - Advanced Social Data Analytics

September 24, 2026

Name: [Your name]

PID: [Your PID]

Work **individually** during class. Complete any remaining work and submit your **PDF to Gradescope by Friday, September 25, at 11:59 p.m. (Pacific time)**.

Practice basic R commands from DSS Chapter 1 (pp. 8-25) and render this Quarto (.qmd) document to PDF.

Part 1: Get started with Quarto

Exercise 1.1: Get started

Open lab00.qmd in RStudio's **Source** editor on [Datahub](#). Enter your name and PID above, save, and click **Render**. Find the following code and its result in the PDF.

```
1 + 3
```

```
[1] 4
```

If setup or rendering fails, ask the instructor or PLA for help during class.

Where to write code and answers

A QMD file combines **R code chunks** and **Markdown text**. In Source, find the chunk around `1 + 3` above: it opens with three backticks and an `r` in braces, and closes with three backticks.

- **Code:** Inside each exercise chunk, replace `# YOUR CODE HERE` with R commands. Change `#| eval: false` to `#| eval: true` when the chunk is complete. Complete and enable earlier chunks first. Keep backticks and `#| label:` lines unchanged.
- **Written answers:** Outside chunks, replace `[Your answer here]` with ordinary sentences. Markdown adds formatting: **`**bold**`** makes text bold; **`##`** Heading creates a heading.

Leave the document settings at the top unchanged. **Render** runs enabled chunks from top to bottom and combines your text, code, and output in a PDF.

Help: [Quarto in RStudio](#) and [Markdown basics](#).

Part 2: Use R as a calculator

R uses `+`, `-`, `*`, and `/` for arithmetic. Parentheses control which calculation happens first. Inside an R chunk, `#` starts a comment for R. Lines beginning with `#|` also supply settings for rendering.

Keep chunk labels: `#| label: exercise-2-1-calculator` identifies Exercise 2.1 in render progress and R-code error messages. **Do not delete or rename label lines;** they help us locate problems when rendering fails.

Exercise 2.1: Calculate and compare

Use R to calculate both expressions below, on separate lines:

- $(4 - 1 + 3) * (2 / 3)$
- $4 - 1 + 3 * 2 / 3$

```
# YOUR CODE HERE
```

Question: Why do the expressions give different results? Use one sentence.

[Your answer here]

Exercise 2.2: Create a vector and find its mean

`<-` stores a value in an object. `c()` combines values into a vector, and `mean()` calculates their average. The supplied line stores three values in a vector named `x`.

Run that line, then add code to **print `x`** and **calculate its mean**. To print an object, write its name on a line by itself; to find its mean, put its name inside `mean()`.

```
x <- c(10, 8, 12)
# YOUR CODE HERE
```

Question: What is the mean for these three values?

[Your answer here]

Part 3: Load election data

`elections.csv` contains U.S. presidential election results. Each row records the votes in one county in one election year. Keep the CSV and QMD in the same folder.

Variable	Meaning
<code>year</code>	Election year
<code>state, county</code>	State and county names
<code>rep</code>	Votes for the Republican candidate
<code>dem</code>	Votes for the Democratic candidate
<code>other</code>	Votes for other candidates

Exercise 3.1: Read and preview the data

Load `elections.csv` using `read.csv()` and store it in an object named `elections`. Then use `head()` to print its first six rows.

Hint: `object <- read.csv("filename.csv")` loads a file; `head(object)` previews it. Keep the CSV and QMD in the same folder.

```
# YOUR CODE HERE
```

Exercise 3.2: Compute and interpret a mean

Use `mean()` to find the average number of Democratic votes (`dem`) across **all rows** of `elections`. The expression `elections$dem` selects that column. Complete and enable Exercise 3.1 first.

```
# YOUR CODE HERE
```

Question: In one sentence, report the average number of Democratic votes per county per election in this dataset.

[Your answer here]

Part 4: Render and submit

1. Save your work.
2. Click **Render**. Open the PDF and check your name, code, output, and written answers.
3. In **Files**, select `lab00.pdf`, then **More > Export** to download it. Submit that PDF to Gradescope by **Friday at 11:59 p.m.** Keep your `.qmd` for future edits.